

Exam Formula Sheet

Epi 202: Probability

$$\begin{aligned}\text{Var}(\tilde{a} \cdot \tilde{X}) &= \text{Var}\left(\sum_{i=1}^n a_i X_i\right) \\ &= \tilde{a}^\top \text{Var}(\tilde{X}) \tilde{a} \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j)\end{aligned}$$

$$\text{E}[Y] = \text{E}[\text{E}[Y \mid X]]$$

$$\text{E}[Y \mid Z] = \text{E}[\text{E}[Y \mid X, Z] \mid Z]$$

$$\text{Var}(Y) = \text{E}[\text{Var}(Y \mid X)] + \text{Var}(\text{E}[Y \mid X])$$

$$\text{Cov}(Y, Z) = \text{E}[\text{Cov}(Y, Z \mid X)] + \text{Cov}(\text{E}[Y \mid X], \text{E}[Z \mid X])$$

Epi 203: Statistical inference

$$\mathcal{L}(\theta) \stackrel{\text{def}}{=} \text{p}(\tilde{X} = \tilde{x} \mid \Theta = \theta)$$

$$\ell \stackrel{\text{def}}{=} \log\{\mathcal{L}(\tilde{x} \mid \theta)\}$$

$$\ell' \stackrel{\text{def}}{=} \frac{\partial}{\partial \theta} \ell(\tilde{x} \mid \theta)$$

$$\ell'' \stackrel{\text{def}}{=} \frac{\partial}{\partial \tilde{\theta}} \frac{\partial}{\partial \tilde{\theta}^\top} \ell(\tilde{x} \mid \tilde{\theta})$$

$$\ell''_{ij} = \frac{\partial}{\partial \theta_i} \frac{\partial}{\partial \theta_j} \ell(\tilde{X} = \tilde{x} \mid \tilde{\theta})$$

$$I \stackrel{\text{def}}{=} -\ell''(\tilde{x} \mid \tilde{\theta})$$

$$\mathcal{I} \stackrel{\text{def}}{=} \text{E}[I(\tilde{x} \mid \theta)]$$

$$\hat{\theta}_{ML} \sim \text{N}(\theta, [\mathcal{I}(\tilde{\theta})]^{-1})$$

For one parameter θ_k :

$$\text{CI}_{1-\alpha}(\theta_k) = [\hat{\theta}_k \pm z_{1-\frac{\alpha}{2}} \widehat{\text{SE}}(\hat{\theta}_k)]$$

$$Z_k \stackrel{\text{def}}{=} \frac{\hat{\theta}_k - \theta_{k,0}}{\widehat{\text{SE}}(\hat{\theta}_k)} \sim \text{N}(0, 1) \quad \text{under } H_0 : \theta_k = \theta_{k,0} \quad z_k = \text{observed } Z_k$$

$$\begin{aligned}p\text{-value} &= 2 \text{Pr}(|Z| \geq |z_k|), \quad Z \sim \text{N}(0, 1) \\ &= 2(1 - \Phi(|z_k|))\end{aligned}$$

Sta 108: Linear regression

$$t_k \stackrel{\text{def}}{=} \frac{\hat{\beta}_k - \beta_{k,0}}{\widehat{\text{SE}}(\hat{\beta}_k)} \sim t_{n-p} \quad \text{under } H_0 : \beta_k = \beta_{k,0} \quad t_k^{\text{obs}} = \text{observed } t_k$$

$$\text{CI}_{1-\alpha}(\beta_k) = \left[\hat{\beta}_k \pm t_{n-p} \left(1 - \frac{\alpha}{2} \right) \widehat{\text{SE}}(\hat{\beta}_k) \right]$$

Let $T_{n-p} \sim t_{n-p}$.

$$\begin{aligned} p\text{-value} &= 2 \Pr(|T_{n-p}| \geq |t_k^{\text{obs}}|) \\ &= 2(1 - F_{t_{n-p}}(|t_k^{\text{obs}}|)) \end{aligned}$$

$$\text{CI}_{1-\alpha}(\mu(\tilde{x}^*)) = \left[\hat{\mu}(\tilde{x}^*) \pm t_{n-p} \left(1 - \frac{\alpha}{2} \right) \widehat{\text{SE}}(\hat{\mu}(\tilde{x}^*)) \right]$$

Y^* denotes a new observation (not in the training data), with corresponding covariate pattern $\tilde{x}^*$. Let $\hat{Y}^* \stackrel{\text{def}}{=} \hat{\mu}(\tilde{x}^*)$.

$$\text{PI}_{1-\alpha}(Y^*|\tilde{x}^*) = \left[\hat{Y}^* \pm t_{n-p} \left(1 - \frac{\alpha}{2} \right) \widehat{\text{SE}}(Y^* - \hat{Y}^*) \right]$$

$$\widehat{\text{SE}}(Y^* - \hat{Y}^*) = \hat{\sigma} \sqrt{1 + (\tilde{x}^*)^\top (\mathbf{X}'\mathbf{X})^{-1} \tilde{x}^*}$$

$$\text{Var}(Y^* - \hat{Y}^*) = \sigma^2 (1 + (\tilde{x}^*)^\top (\mathbf{X}'\mathbf{X})^{-1} \tilde{x}^*)$$

Let $\hat{\Sigma} \stackrel{\text{def}}{=} \widehat{\text{Var}}(\hat{\beta}) = \hat{\sigma}^2 (\mathbf{X}'\mathbf{X})^{-1}$.

$$\widehat{\text{Var}}(\hat{\mu}(\tilde{x})) = \tilde{x}^\top \hat{\Sigma} \tilde{x}$$

Let $\Delta\mu(\tilde{x}, \tilde{x}^*) = \mu(\tilde{x}) - \mu(\tilde{x}^*)$, and let $\Delta\tilde{x} = \tilde{x} - \tilde{x}^*$; then:

$$\widehat{\text{Var}}(\widehat{\Delta\mu}(\tilde{x}, \tilde{x}^*)) = \Delta\tilde{x}^\top \hat{\Sigma} \Delta\tilde{x}$$

Epi 204: Generalized linear models

Generalized linear models have three components:

1. The **outcome distribution** family: $p(Y|\mu(\tilde{x}))$
2. The **link function**: $g(\mu(\tilde{x})) = \eta(\tilde{x})$
3. The **linear component**: $\eta(\tilde{x}) = \tilde{x} \cdot \beta$

$$\theta_\omega(\tilde{x}, \tilde{x}^*) = \exp\{(\Delta\tilde{x}) \cdot \tilde{\beta}\}$$

Estimates of odds ratios from 2x2 contingency tables

$$\hat{\theta} = \frac{ad}{bc}$$

Summary of logistic regression definitions and results

Odds and log-odds

These association-measure results are covered in Measures of Association for Binary Outcomes¹.

Odds (def²)

$$\omega \stackrel{\text{def}}{=} \frac{\Pr(A)}{\Pr(\neg A)}$$

Conditional odds (def³)

$$\omega(A|B) \stackrel{\text{def}}{=} \frac{\Pr(A|B)}{\Pr(\neg A|B)}$$

Odds function (def⁴)

$$\text{odds}\{\pi\} \stackrel{\text{def}}{=} \frac{\pi}{1 - \pi}$$

¹binary-outcome-associations.qmd

²binary-outcome-associations.qmd#def-odds

³binary-outcome-associations.qmd#def-c-odds

⁴binary-outcome-associations.qmd#def-odds-fn

Probability to odds (thm⁵, cor⁶)

$$\omega = \frac{\pi}{1 - \pi} = \text{odds}\{\pi\}$$

Simplified odds expressions; odds of a non-event (thm⁷, cor⁸)

$$\text{odds}\{\pi\} = \frac{1}{\pi^{-1} - 1} = (\pi^{-1} - 1)^{-1}, \quad \omega(\neg A) = \frac{1 - \pi}{\pi} = \pi^{-1} - 1$$

Odds ratio (def⁹)

$$\theta(\omega_1, \omega_2) \stackrel{\text{def}}{=} \frac{\omega_1}{\omega_2}$$

OR as ratio of probability ratios (thm¹⁰)

$$\theta(\omega_1, \omega_2) = \frac{\omega_1}{\omega_2} = \frac{\left(\frac{\pi_1}{1 - \pi_1}\right)}{\left(\frac{\pi_2}{1 - \pi_2}\right)}$$

Odds ratios are reversible, marginally and conditionally (thm¹¹, thm¹²)

$$\theta_\omega(A|B) = \theta_\omega(B|A), \quad \theta_\omega(A|B, C) = \theta_\omega(B|A, C)$$

Inverse-odds and probability recovery

Odds to probability (thm¹³)

$$\pi = \frac{\omega}{1 + \omega}$$

Inverse-odds function (def¹⁴; recovers probability by cor¹⁵)

$$\text{invodds}\{\omega\} \stackrel{\text{def}}{=} \frac{\omega}{1 + \omega}, \quad \pi = \text{invodds}\{\omega\}$$

Simplified inverse-odds (lem¹⁶)

$$\text{invodds}\{\omega\} = \frac{1}{1 + \omega^{-1}} = (1 + \omega^{-1})^{-1}$$

One minus inverse-odds, and its complement (lem¹⁷, cor¹⁸)

$$1 - \pi = \frac{1}{1 + \omega}, \quad 1 + \omega = \frac{1}{1 - \pi}$$

Log-odds (logit) and expit

Log-odds, and log-odds from probability (def¹⁹, thm²⁰)

$$\eta \stackrel{\text{def}}{=} \log\{\omega\}, \quad \eta = \log\left\{\frac{\pi}{1 - \pi}\right\}$$

Logit function, and expanded form (def²¹, thm²²)

$$\text{logit}(\pi) \stackrel{\text{def}}{=} \log\{\text{odds}\{\pi\}\} = \log\left\{\frac{\pi}{1 - \pi}\right\}$$

⁵[binary-outcome-associations.qmd#thm-prob-to-odds](#)

⁶[binary-outcome-associations.qmd#cor-oddsf-to-odds](#)

⁷[binary-outcome-associations.qmd#thm-odds-simplified](#)

⁸[binary-outcome-associations.qmd#cor-odds-neg](#)

⁹[binary-outcome-associations.qmd#def-OR](#)

¹⁰[binary-outcome-associations.qmd#thm-or-ratio-ratio](#)

¹¹[binary-outcome-associations.qmd#thm-or-swap](#)

¹²[binary-outcome-associations.qmd#thm-conditional-OR-swap](#)

¹³[binary-outcome-associations.qmd#thm-odds-to-prob](#)

¹⁴[binary-outcome-associations.qmd#def-inv-odds](#)

¹⁵[binary-outcome-associations.qmd#cor-invodds-pi](#)

¹⁶[binary-outcome-associations.qmd#lem-invodds-simplified](#)

¹⁷[binary-outcome-associations.qmd#lem-one-minus-odds-inv](#)

¹⁸[binary-outcome-associations.qmd#cor-inverse-odds-nonevent](#)

¹⁹[binary-outcome-associations.qmd#def-logodds](#)

²⁰[binary-outcome-associations.qmd#thm-logodds-pi](#)

²¹[binary-outcome-associations.qmd#def-logit-fn](#)

²²[binary-outcome-associations.qmd#thm-logit-function](#)

Log-odds equals logit (cor²³)

$$\eta = \text{logit}\{\pi\}$$

Odds and probability from log-odds (lem²⁴, thm²⁵)

$$\omega = \exp\{\eta\}, \quad \pi = \frac{\exp\{\eta\}}{1 + \exp\{\eta\}}$$

Expit / inverse-logit function, and equivalent expressions (def²⁶, thm²⁷)

$$\text{expit}(\eta) \stackrel{\text{def}}{=} \text{invodds}\{\exp\{\eta\}\} = \frac{\exp\{\eta\}}{1 + \exp\{\eta\}} = (1 + \exp\{-\eta\})^{-1}$$

Probability as expit (thm²⁸)

$$\pi = \text{expit}\{\eta\} \tag{1}$$

Logit and expit are inverses (thm²⁹)

$$\text{logit}\{\text{expit}\{\eta\}\} = \eta \quad \text{expit}\{\text{logit}\{\pi\}\} = \pi$$

$$\left[\pi \stackrel{\text{def}}{=} \Pr(Y = 1 | \tilde{X} = \tilde{x}) \right] \xleftrightarrow[\underbrace{\frac{\omega}{1+\omega}}_{\text{expit}(\eta)}]{\underbrace{\frac{\pi}{1-\pi}}_{\text{logit}(\pi)}} \left[\omega \stackrel{\text{def}}{=} \text{odds}\{Y = 1 | \tilde{X} = \tilde{x}\} \right] \xleftrightarrow[\underbrace{\exp\{\eta\}}_{\text{log}\{\omega\}}]{\frac{\log\{\omega\}}{\exp\{\eta\}}} \left[\eta(\tilde{x}) \stackrel{\text{def}}{=} \eta(Y = 1 | \tilde{X} = \tilde{x}) \right]$$

Figure 1: Diagram of logistic regression link and inverse link functions

Rare events

Odds minus probability (thm³⁰)

$$\omega - \pi = \frac{\pi^2}{1 - \pi}, \quad \text{where } \omega = \frac{\pi}{1 - \pi}$$

OR as RR times a correction factor (thm³¹)

$$\theta(\omega_1, \omega_2) = \rho(\pi_1, \pi_2) \cdot \frac{1 - \pi_2}{1 - \pi_1}$$

When π_1 and π_2 are both small, the correction factor is close to 1, so the odds ratio is close to the risk ratio.

Derivatives

	π	ω	η	$\tilde{x}$	$\tilde{\beta}$
π	1	$(1 + \omega)^2$	$\frac{(1 + \omega)^2}{\omega}$	undef	undef
ω	$(1 - \pi)^2$	1	$\frac{1}{\omega}$	undef	undef
η	$\pi(1 - \pi)$	ω	1	undef	undef
$\tilde{x}$	$\underbrace{\tilde{\beta}\pi(1 - \pi)}_{p \times 1}$	$\underbrace{\tilde{\beta}\omega}_{p \times 1}$	$\underbrace{\tilde{\beta}}_{p \times 1}$	$\underbrace{\mathbb{I}}_{p \times p}$	$\mathbf{0}_{p \times p}$
$\tilde{\beta}$	$\underbrace{\tilde{x}\pi(1 - \pi)}_{p \times 1}$	$\underbrace{\tilde{x}\omega}_{p \times 1}$	$\underbrace{\tilde{x}}_{p \times 1}$	$\mathbf{0}_{p \times p}$	$\underbrace{\mathbb{I}}_{p \times p}$

²³binary-outcome-associations.qmd#cor-logodds-logit

²⁴binary-outcome-associations.qmd#lem-odds-from-logodds

²⁵binary-outcome-associations.qmd#thm-prob-from-logodds

²⁶binary-outcome-associations.qmd#def-expit

²⁷binary-outcome-associations.qmd#thm-expit-expressions

²⁸binary-outcome-associations.qmd#thm-expit-prob-logodds

²⁹binary-outcome-associations.qmd#thm-logit-expit

³⁰binary-outcome-associations.qmd#thm-odds-minus-probs

³¹binary-outcome-associations.qmd#thm-rare-disease-or-rr

Column labels indicate the numerators of the derivatives; row labels indicate the denominators (dimensions and details in tbl³²).

Derivative of expit (cor³³)

$$\text{expit}'\{\eta\} = \text{expit}\{\eta\}(1 - \text{expit}\{\eta\})$$

Logistic regression model

Logistic regression (def³⁴)

$$Y_i | \tilde{X}_i \sim_{\perp} \text{Ber}(\pi(\tilde{X}_i)), \quad \text{logit}\{\pi(\tilde{x})\} = \tilde{x}^\top \tilde{\beta}$$

Throughout this summary, $\tilde{\beta} = (\beta_0, \beta_1, \dots, \beta_p)$ is the $(p+1) \times 1$ coefficient vector (including the intercept), $\mathbf{X}$ is the $n \times (p+1)$ design matrix whose i^{th} row is $\tilde{x}_i^\top$, and $\tilde{\varepsilon}$ is the $n \times 1$ error vector with components $\varepsilon_i \stackrel{\text{def}}{=} y_i - \pi_i$.

Log-likelihood and score function

Log-likelihood component (lem³⁵)

$$\ell_i(\pi_i) = y_i \eta_i - \log\{1 + \omega_i\}$$

Score function as sum of components (lem³⁶, thm³⁷)

$$\tilde{\ell}'(\tilde{\beta}) = \sum_{i=1}^n \tilde{\ell}'_i(\tilde{\beta}), \quad \ell'_i(\tilde{\beta}) = \tilde{x}_i \varepsilon_i$$

Score function (thm³⁸)

$$\tilde{\ell}'(\tilde{\beta}) = \sum_{i=1}^n \tilde{x}_i \varepsilon_i = \underbrace{\mathbf{X}^\top}_{(p+1) \times n} \underbrace{\tilde{\varepsilon}}_{n \times 1}$$

Maximum likelihood estimation

Estimating equation (eq³⁹) The MLE $\hat{\tilde{\beta}}$ solves the score equation:

$$\underbrace{\tilde{\ell}'(\hat{\tilde{\beta}})}_{(p+1) \times 1} = \sum_{i=1}^n \underbrace{\tilde{x}_i}_{(p+1) \times 1} \underbrace{\left(y_i - \text{expit}\left\{ \tilde{x}_i^\top \hat{\tilde{\beta}} \right\} \right)}_{1 \times 1} = \mathbf{0}_{(p+1) \times 1}$$

This estimating equation has no closed-form solution in general; $\hat{\tilde{\beta}}$ must be computed by numerical iteration.

Hessian (eq⁴⁰)

$$\underbrace{\ell''(\tilde{\beta})}_{(p+1) \times (p+1)} = - \underbrace{\mathbf{X}^\top}_{(p+1) \times n} \underbrace{\mathbf{D}}_{n \times n} \underbrace{\mathbf{X}}_{n \times (p+1)}, \quad \mathbf{D} \stackrel{\text{def}}{=} \text{diag}(\text{Var}(Y_i | \tilde{X}_i = \tilde{x}_i)) = \text{diag}(\pi_i(1 - \pi_i))$$

Concavity (rem⁴¹): the Hessian is negative semi-definite for every $\tilde{\beta}$ (negative definite when $\mathbf{X}$ has full column rank), so the log-likelihood is concave and every solution of the score equation is a global maximum.

Newton-Raphson update (Newton-Raphson method⁴²)

$$\underbrace{\hat{\tilde{\beta}}^*}_{(p+1) \times 1} \leftarrow \underbrace{\hat{\tilde{\beta}}}_{(p+1) \times 1} - \underbrace{\left(\ell''(\tilde{y}; \hat{\tilde{\beta}}^*) \right)^{-1}}_{(p+1) \times (p+1)} \underbrace{\ell'(\tilde{y}; \hat{\tilde{\beta}}^*)}_{(p+1) \times 1}$$

One-sample MLE for odds (thm⁴³)

$$\hat{\omega} = \frac{x}{n - x}$$

³²logistic-regression.qmd#tbl-logistic-deriv-matrix

³³logistic-regression.qmd#cor-deriv-expit

³⁴logistic-regression.qmd#def-logistic-regression

³⁵logistic-regression.qmd#lem-logistic-loglik-component

³⁶logistic-regression.qmd#lem-score-logistic

³⁷logistic-regression.qmd#thm-logistic-score-comp

³⁸logistic-regression.qmd#thm-logistic-score-fn

³⁹logistic-regression.qmd#eq-score-logistic

⁴⁰logistic-regression.qmd#eq-logistic-hess

⁴¹logistic-regression.qmd#rem-hessian-nsd

⁴²intro-MLEs.qmd#sec-newton-raphson

⁴³binary-outcome-associations.qmd#thm-est-odds

Wald tests and confidence intervals for coefficients

Wald test statistic for $H_0 : \beta_k = \beta_{k,0}$ (CLT for MLEs⁴⁴)

$$z_k = \frac{\hat{\beta}_k - \beta_{k,0}}{\widehat{\text{SE}}(\hat{\beta}_k)}, \quad z_k \sim N(0,1) \text{ under } H_0$$

95% confidence intervals for β_k and e^{β_k} (MLE invariance)

$$\hat{\beta}_k \pm 1.96 \cdot \widehat{\text{SE}}(\hat{\beta}_k), \quad e^{\beta_k} \in \left(e^{\hat{\beta}_k - 1.96 \cdot \widehat{\text{SE}}(\hat{\beta}_k)}, e^{\hat{\beta}_k + 1.96 \cdot \widehat{\text{SE}}(\hat{\beta}_k)} \right)$$

For $k \neq 0$, e^{β_k} is an odds ratio; for $k = 0$, e^{β_0} is the baseline odds.

Odds ratios in logistic regression

General OR formula (lem⁴⁵)

$$\theta_\omega(\tilde{x}, \tilde{x}^*) = \exp\{\eta(\tilde{x}) - \eta(\tilde{x}^*)\}$$

Difference in log-odds; OR in terms of $\Delta\eta$ (def⁴⁶, cor⁴⁷)

$$\Delta\eta \stackrel{\text{def}}{=} \eta(\tilde{x}) - \eta(\tilde{x}^*), \quad \theta_\omega(\tilde{x}, \tilde{x}^*) = \exp\{\Delta\eta\}$$

$\Delta\eta$ from covariates (lem⁴⁸)

$$\Delta\eta = (\tilde{x} - \tilde{x}^*) \cdot \tilde{\beta}$$

Difference in covariate patterns; $\Delta\eta$ from $\Delta\tilde{x}$ (def⁴⁹, cor⁵⁰)

$$\Delta\tilde{x} \stackrel{\text{def}}{=} \tilde{x} - \tilde{x}^*, \quad \Delta\eta = \Delta\tilde{x} \cdot \tilde{\beta}$$

OR in terms of $\Delta\tilde{x}$ (thm⁵¹)

$$\theta_\omega(\tilde{x}, \tilde{x}^*) = \exp\{(\Delta\tilde{x}) \cdot \tilde{\beta}\}$$

Log OR equals $\Delta\eta$ (cor⁵²)

$$\log\{\theta_\omega(\tilde{x}, \tilde{x}^*)\} = \Delta\eta$$

Inference for log-odds and odds ratios

Estimated SE of log-odds (thm⁵³)

$$\widehat{\text{Var}}(\hat{\eta}(\tilde{x})) = \underbrace{\tilde{x}^\top}_{1 \times (p+1)} \underbrace{\hat{\Sigma}}_{(p+1) \times (p+1)} \underbrace{\tilde{x}}_{(p+1) \times 1}, \quad \widehat{\text{SE}}(\hat{\eta}(\tilde{x})) = \sqrt{\tilde{x}^\top \hat{\Sigma} \tilde{x}}$$

Estimated SE of $\Delta\hat{\eta}$ (thm⁵⁴)

$$\widehat{\text{Var}}(\Delta\hat{\eta}) = \underbrace{\Delta\tilde{x}^\top}_{1 \times (p+1)} \underbrace{\hat{\Sigma}}_{(p+1) \times (p+1)} \underbrace{(\Delta\tilde{x})}_{(p+1) \times 1}, \quad \widehat{\text{SE}}(\Delta\hat{\eta}) = \sqrt{\Delta\tilde{x}^\top \hat{\Sigma} (\Delta\tilde{x})}$$

CI for an odds ratio (eq⁵⁵; details in sec⁵⁶): exponentiate the endpoints of the CI for $\Delta\eta$:

$$\theta_\omega(\tilde{x}, \tilde{x}^*) \in (e^L, e^R), \quad (L, R) = \widehat{\Delta\eta} \mp 1.96 \cdot \widehat{\text{SE}}(\widehat{\Delta\eta})$$

⁴⁴intro-MLEs.qmd#thm-dist-mle

⁴⁵logistic-regression.qmd#lem-logistic-OR

⁴⁶logistic-regression.qmd#def-difflogodds

⁴⁷logistic-regression.qmd#cor-logistic-OR

⁴⁸logistic-regression.qmd#lem-diff-logodds

⁴⁹logistic-regression.qmd#def-diff-covs

⁵⁰logistic-regression.qmd#cor-diff-logodds

⁵¹logistic-regression.qmd#thm-logistic-OR

⁵²logistic-regression.qmd#cor-log-or

⁵³logistic-regression.qmd#thm-est-se-logodds

⁵⁴logistic-regression.qmd#thm-est-se-diff-logodds

⁵⁵logistic-regression.qmd#eq-ci-OR-exp

⁵⁶logistic-regression.qmd#sec-or-inference

Odds ratios in study designs

Disease and exposure odds ratios (def⁵⁷, def⁵⁸)

$$\theta_{\omega}(D|E) \stackrel{\text{def}}{=} \frac{\omega(D|E)}{\omega(D|\neg E)}, \quad \theta_{\omega}(E|D) \stackrel{\text{def}}{=} \frac{\omega(E|D)}{\omega(E|\neg D)}$$

Case-control studies (section⁵⁹): outcome-stratified sampling breaks estimation of risks, risk ratios, and the components of the disease odds ratio (def⁶⁰), but the exposure odds ratio (def⁶¹) is estimable, and it equals the disease odds ratio by reversibility (thm⁶²).

2×2 table shortcut (table⁶³; cell counts a, b = exposed events and non-events, c, d = unexposed events and non-events):

$$\hat{\omega}(\text{Event}|\text{Exposed}) = \frac{a}{b}, \quad \hat{\omega}(\text{Event}|\neg \text{Exposed}) = \frac{c}{d}, \quad \theta_{\omega}(\text{Exposed}, \neg \text{Exposed}) = \frac{ad}{bc}$$

Model fit and comparison

Details in sec⁶⁴; here q is the number of distinct covariate patterns, p is the number of β parameters, and ℓ_{full} is the saturated-model log-likelihood.

Deviance test (def⁶⁵)

$$D \stackrel{\text{def}}{=} 2(\ell_{\text{full}} - \ell(\hat{\beta}; \mathbf{x})), \quad D \sim \chi^2(q - p)$$

Hosmer-Lemeshow test (def⁶⁶; g groups by predicted probability)

$$X^2 \stackrel{\text{def}}{=} \sum_{c=1}^g \frac{(o_c - e_c)^2}{e_c} \sim \chi^2(g - 2)$$

Pearson chi-squared GOF test (sum of squared Pearson residuals, def⁶⁷)

$$X^2 = \sum_{k=1}^q X_k^2 \sim \chi^2(q - p)$$

AIC and BIC [lower is better] (AIC⁶⁸ (Akaike 1974); BIC⁶⁹ (Schwarz 1978))

$$\text{AIC} = -2\ell(\hat{\theta}) + 2p, \quad \text{BIC} = -2\ell(\hat{\theta}) + p \log\{n\}$$

Likelihood ratio test (def⁷⁰; for nested models with $p_0 < p_1$ parameters — here ℓ_1 and ℓ_0 are the two fitted models' log-likelihoods, not the saturated model's)

$$\Lambda \stackrel{\text{def}}{=} 2(\ell_1(\hat{\theta}_1) - \ell_0(\hat{\theta}_0)), \quad \Lambda \sim \chi^2(p_1 - p_0) \text{ under } H_0$$

by Wilks' theorem⁷¹.

Residual-based diagnostics

Residuals are computed per covariate pattern k (grouped data); h_k denotes the leverage of pattern k (def⁷²).

Response residual (def⁷³)

$$e_k \stackrel{\text{def}}{=} \bar{y}_k - \hat{\pi}(x_k)$$

Pearson residual (and standardized version) (def⁷⁴, def⁷⁵)

$$X_k \stackrel{\text{def}}{=} \frac{\bar{y}_k - \hat{\pi}_k}{\sqrt{\hat{\pi}_k(1 - \hat{\pi}_k)/n_k}}, \quad r_{P_k} \stackrel{\text{def}}{=} \frac{X_k}{\sqrt{1 - h_k}}$$

⁵⁷binary-outcome-associations.qmd#def-disease-OR

⁵⁸binary-outcome-associations.qmd#def-exposure-OR

⁵⁹binary-outcome-associations.qmd#sec-CC-studies

⁶⁰binary-outcome-associations.qmd#def-disease-OR

⁶¹binary-outcome-associations.qmd#def-exposure-OR

⁶²binary-outcome-associations.qmd#thm-or-swap

⁶³binary-outcome-associations.qmd#tbl-2x2-generic

⁶⁴logistic-regression.qmd#sec-gof

⁶⁵logistic-regression.qmd#def-deviance-test

⁶⁶logistic-regression.qmd#def-hosmer-lemeshow

⁶⁷logistic-regression.qmd#def-pearson-residual

⁶⁸Linear-models-overview.qmd#def-aic

⁶⁹Linear-models-overview.qmd#def-bic

⁷⁰logistic-regression.qmd#def-lrt

⁷¹intro-MLEs.qmd#thm-wilks

⁷²logistic-regression.qmd#def-leverage-glm

⁷³logistic-regression.qmd#def-response-residual-logistic

⁷⁴logistic-regression.qmd#def-pearson-residual

⁷⁵logistic-regression.qmd#def-standardized-pearson-residual

Deviance residual (and standardized version) (def⁷⁶)

$$d_k \stackrel{\text{def}}{=} \text{sign}(y_k - n_k \hat{\pi}_k) \left\{ \sqrt{2[\ell_{\text{full}}(x_k) - \ell(\hat{\beta}; x_k)]} \right\}, \quad r_{D_k} \stackrel{\text{def}}{=} \frac{d_k}{\sqrt{1 - h_k}}$$

Cook's distance (def⁷⁷)

$$D_k \stackrel{\text{def}}{=} \frac{(r_{D_k})^2}{p} \cdot \frac{h_k}{1 - h_k}$$

Leverage (def⁷⁸): h_k is the k -th diagonal element of the weighted hat matrix (with $\mathbf{X}$ here the $q \times p$ design matrix of covariate patterns):

$$\mathbf{H} \stackrel{\text{def}}{=} \underbrace{\mathbf{W}^{1/2}}_{q \times q} \underbrace{\mathbf{X}}_{q \times p} \left(\underbrace{\mathbf{X}^\top \mathbf{W} \mathbf{X}}_{p \times p} \right)^{-1} \underbrace{\mathbf{X}^\top}_{p \times q} \underbrace{\mathbf{W}^{1/2}}_{q \times q}, \quad \mathbf{W} \stackrel{\text{def}}{=} \text{diag}(n_k \hat{\pi}_k (1 - \hat{\pi}_k))_{q \times q}$$

Comparing probabilities

Risk difference (def⁷⁹)

$$\delta(\pi_1, \pi_2) \stackrel{\text{def}}{=} \pi_1 - \pi_2$$

Risk ratio (def⁸⁰)

$$\rho(\pi_1, \pi_2) \stackrel{\text{def}}{=} \frac{\pi_1}{\pi_2}$$

Relative risk difference (def⁸¹; equals RR minus 1 by thm⁸²)

$$\xi(\pi_1, \pi_2) \stackrel{\text{def}}{=} \frac{\delta(\pi_1, \pi_2)}{\pi_2} = \rho(\pi_1, \pi_2) - 1$$

Marginal risks from logistic regression

Conditional predicted risk (def⁸³)

$$\pi(a, z) \stackrel{\text{def}}{=} \Pr(Y = 1 \mid A = a, Z = z)$$

Conditional RD and RR (def⁸⁴, def⁸⁵)

$$\text{RD}(z) \stackrel{\text{def}}{=} \pi(1, z) - \pi(0, z), \quad \text{RR}(z) \stackrel{\text{def}}{=} \frac{\pi(1, z)}{\pi(0, z)}$$

Observed marginal risk (def⁸⁶; standardization form by thm⁸⁷)

$$\pi(a) \stackrel{\text{def}}{=} \Pr(Y = 1 \mid A = a) = \mathbb{E}[\pi(a, Z) \mid A = a]$$

Potential mean and causal marginal risk (def⁸⁸, def⁸⁹; identification form by thm⁹⁰)

$$\mu_a \stackrel{\text{def}}{=} \mathbb{E}[Y^a], \quad \pi_a \stackrel{\text{def}}{=} \Pr(Y^a = 1) = \mathbb{E}[\pi(a, Z)]$$

No confounding: observed = causal (cor⁹¹; the two estimands are contrasted in rem⁹²)

$$Z \perp\!\!\!\perp A \implies \pi(a) = \pi_a$$

⁷⁶logistic-regression.qmd#def-deviance-residual-logistic

⁷⁷logistic-regression.qmd#def-cooks-distance-glm

⁷⁸logistic-regression.qmd#def-leverage-glm

⁷⁹binary-outcome-associations.qmd#def-RD

⁸⁰binary-outcome-associations.qmd#def-RR

⁸¹binary-outcome-associations.qmd#def-RRD

⁸²binary-outcome-associations.qmd#thm-rrd-vs-rr

⁸³logistic-regression.qmd#def-cond-predicted-risk

⁸⁴logistic-regression.qmd#def-cond-rd

⁸⁵logistic-regression.qmd#def-cond-rrr

⁸⁶logistic-regression.qmd#def-observed-marginal-risk

⁸⁷logistic-regression.qmd#thm-observed-marginal-risk

⁸⁸logistic-regression.qmd#def-potential-mean

⁸⁹logistic-regression.qmd#def-causal-marginal-risk

⁹⁰logistic-regression.qmd#thm-causal-marginal-risk

⁹¹logistic-regression.qmd#cor-observed-equals-causal

⁹²logistic-regression.qmd#rem-observed-vs-causal-marginal-risk

G-computation estimator (def⁹³; consistent for π_a by thm⁹⁴)

$$\hat{\pi}_a \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^n \hat{\pi}_{a,i}, \quad \hat{\pi}_{a,i} \stackrel{\text{def}}{=} \hat{\pi}(a, z_i)$$

Predictive margins (def⁹⁵; full-sample form def⁹⁶ is consistent for π_a by cor⁹⁷)

$$\widehat{\text{PM}}(a \mid S) \stackrel{\text{def}}{=} \frac{1}{|S|} \sum_{i \in S} \hat{\pi}_{a,i}, \quad \widehat{\text{PM}}(a) \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^n \hat{\pi}_{a,i} = \hat{\pi}_a$$

Exposed-subgroup predictive margin (def⁹⁸; consistent for $\pi(a)$ by thm⁹⁹; equals $\bar{Y}_a$ exactly by lem¹⁰⁰)

$$\hat{\pi}_a(a) \stackrel{\text{def}}{=} \frac{1}{n_a} \sum_{i: A_i=a} \hat{\pi}_{a,i} = \bar{Y}_a$$

Observed marginal RD and RR (def¹⁰¹, def¹⁰²)

$$\text{OMRD} \stackrel{\text{def}}{=} \pi(1) - \pi(0), \quad \text{OMRR} \stackrel{\text{def}}{=} \frac{\pi(1)}{\pi(0)}$$

Causal marginal RD and RR (def¹⁰³, def¹⁰⁴)

$$\text{CMRD} \stackrel{\text{def}}{=} \pi_1 - \pi_0, \quad \text{CMRR} \stackrel{\text{def}}{=} \frac{\pi_1}{\pi_0}$$

Bootstrap SE and CI for marginal contrasts (def¹⁰⁵; see Bootstrap Confidence Intervals¹⁰⁶ and exm¹⁰⁷): resample the data with replacement B times; refit the logistic model and recompute the marginal contrast on each resample; the bootstrap SE is the standard deviation of the B estimates, and a 95% percentile CI uses their 2.5th and 97.5th percentiles.

Collapsibility (def¹⁰⁸): an estimand θ is collapsible with respect to Z if

$$\theta \stackrel{\text{def}}{=} E[\theta(Z)]$$

Risk differences are collapsible; risk ratios and odds ratios are not collapsible in general (thm¹⁰⁹; counterexample in exm¹¹⁰).

Survival analysis

Probability distribution functions

Table 1: Probability distribution functions

Name	Symbols	Definition
Probability density function (PDF)	$f(t), p(t)$	$p(T = t)$
Cumulative distribution function (CDF)	$F(t), P(t)$	$P(T \leq t)$
Survival function	$S(t), \bar{F}(t)$	$P(T > t)$
Hazard function	$\lambda(t), h(t)$	$p(T = t \mid T \geq t)$
Cumulative hazard function	$\Lambda(t), H(t)$	$\int_{u=-\infty}^t \lambda(u) du$
Log-hazard function	$\eta(t)$	$\log\{\lambda(t)\}$

⁹³logistic-regression.qmd#def-g-computation

⁹⁴logistic-regression.qmd#thm-g-computation-consistency

⁹⁵logistic-regression.qmd#def-predictive-margins

⁹⁶logistic-regression.qmd#def-full-sample-predictive-margin

⁹⁷logistic-regression.qmd#cor-full-sample-pm

⁹⁸logistic-regression.qmd#def-subgroup-predictive-margin

⁹⁹logistic-regression.qmd#thm-subgroup-pm

¹⁰⁰logistic-regression.qmd#lem-subgroup-pm-proportion

¹⁰¹logistic-regression.qmd#def-observed-marginal-rd

¹⁰²logistic-regression.qmd#def-observed-marginal-rr

¹⁰³logistic-regression.qmd#def-causal-marginal-rd

¹⁰⁴logistic-regression.qmd#def-causal-marginal-rr

¹⁰⁵logistic-regression.qmd#def-bootstrap-marginal-rd

¹⁰⁶basic-statistical-methods.qmd#sec-bootstrap-ci

¹⁰⁷logistic-regression.qmd#exm-wcgs-marginal-rd

¹⁰⁸logistic-regression.qmd#def-collapsibility

¹⁰⁹logistic-regression.qmd#thm-collapsibility

¹¹⁰logistic-regression.qmd#exm-collapsibility

Diagram of survival distribution function relationships

$$\begin{aligned} f(t) &\xleftarrow[\frac{-S'(t)}{S(t)\lambda(t)}]{} S(t) \xleftarrow[\exp\{-\Lambda(t)\}]{} \Lambda(t) \xleftarrow[\int_{u=0}^t \lambda(u)du]{} \lambda(t) \xleftarrow[\exp\{\eta(t)\}]{} \eta(t) \\ f(t) &\xrightarrow[\int_{u=t}^{\infty} f(u)du]{} S(t) \xrightarrow[-\log S(t)]{} \Lambda(t) \xrightarrow[\Lambda'(t)]{} \lambda(t) \xrightarrow[\log\{\lambda(t)\}]{} \eta(t) \end{aligned}$$

Survival likelihood contributions, assuming non-informative censoring

$$\begin{aligned} p(Y = y, D = d) &= [f_T(y)]^d [S_T(y)]^{1-d} \\ &= [\lambda_T(y)]^d [S_T(y)] \end{aligned}$$

Nonparametric time-to-event distribution estimators

$$\begin{aligned} \hat{\lambda}_i &= \frac{d_i}{r_i} \\ \hat{\kappa}_i &= 1 - \hat{\lambda}_i = \frac{r_i - d_i}{r_i} \\ \hat{S}_{\text{KM}}(t) &\stackrel{\text{def}}{=} \prod_{\{i: t_i \leq t\}} \hat{\kappa}_i \\ \hat{\Lambda}_{\text{NA}}(t) &\stackrel{\text{def}}{=} \sum_{\{i: t_i \leq t\}} \hat{\lambda}_i \end{aligned}$$

Proportional hazards model structure

Formula	Description
$\lambda(t \tilde{x}) = \lambda_0(t) \cdot \theta_\lambda(\tilde{x})$	Proportional hazards assumption
$\Lambda(t \tilde{x}) = \Lambda_0(t) \cdot \theta_\lambda(\tilde{x})$	Cumulative hazard factorization
$\eta(t \tilde{x}) = \eta_0(t) + \Delta\eta(\tilde{x})$	Log-hazard decomposition
$\Delta\eta(\tilde{x}) = \tilde{x} \cdot \tilde{\beta} = \beta_1 x_1 + \dots + \beta_p x_p$	Linear predictor
$\theta_\lambda(\tilde{x}) = \exp\{\Delta\eta(\tilde{x})\}$	Hazard multiplier
$\theta_\lambda(t \tilde{x} : \tilde{x}^*) \stackrel{\text{def}}{=} \frac{\lambda(t \tilde{x})}{\lambda(t \tilde{x}^*)}$	Hazard ratio definition
$\theta_\lambda(t \tilde{x} : \tilde{x}^*) = \exp\{\Delta\eta(\tilde{x}) - \Delta\eta(\tilde{x}^*)\} = \exp\{(\tilde{x} - \tilde{x}^*) \cdot \tilde{\beta}\}$	Hazard ratio formula

Proportional hazards model partial likelihood formula:

Definition 0.1 (Cox PH partial likelihood).

$$\begin{aligned} \mathcal{L}_i^*(\tilde{\beta}) &\stackrel{\text{def}}{=} \frac{\theta_\lambda(\tilde{x}_{(i)})}{\sum_{k \in R(t_i)} \theta_\lambda(\tilde{x}_k)} \\ \mathcal{L}^*(\tilde{\beta}) &\stackrel{\text{def}}{=} \prod_{\{i: d_i=1\}} \mathcal{L}_i^*(\tilde{\beta}) \end{aligned}$$

where, for the distinct event times $t_1 < \dots < t_K$ of the data set:

- (i) is the subject who fails at t_i (unique under the no-ties assumption);
- $R(t_i) \stackrel{\text{def}}{=} \{j : Y_j \geq t_i\}$ is the risk set at t_i (where $Y_j = \min(T_j, C_j)$ is subject j 's observed follow-up time) — the subjects still under observation just before t_i ;
- d_i is the number of events at t_i (so $d_i = 1$ when there are no ties);
- $\theta_\lambda(\tilde{x}) \stackrel{\text{def}}{=} \exp\{\tilde{x}^\top \tilde{\beta}\}$ is the hazard ratio (risk score) for covariate pattern $\tilde{x}$.

Proportional hazards model baseline cumulative hazard estimator:

Definition 0.2 (Breslow estimator of the baseline cumulative hazard).

$$\hat{\Lambda}_0(t) \stackrel{\text{def}}{=} \sum_{t_i \leq t} \frac{d_i}{\sum_{k \in R(t_i)} \theta_\lambda(\tilde{x}_k)}$$

where, for the distinct event times $t_1 < \dots < t_K$ of the data set:

- d_i is the number of events at t_i ;
- $R(t_i) \stackrel{\text{def}}{=} \{j : Y_j \geq t_i\}$ is the risk set at t_i (where $Y_j = \min(T_j, C_j)$ is subject j 's observed follow-up time) — the subjects still under observation just before t_i ;
- $\theta_\lambda(\tilde{x}) \stackrel{\text{def}}{=} \exp\{\tilde{x}^\top \tilde{\beta}\}$ is the hazard ratio (risk score) for covariate pattern $\tilde{x}$.

Exm

Example 0.1 (Breslow estimator with two event times). Suppose there are two distinct event times $t_1 = 2$ and $t_2 = 4$ (no ties, $d_1 = d_2 = 1$), with risk-set-weighted totals $\sum_{k \in R(2)} \theta_\lambda(\tilde{x}_k) = 10$ and $\sum_{k \in R(4)} \theta_\lambda(\tilde{x}_k) = 6$. Then

$$\hat{\Lambda}_0(4) = \frac{1}{10} + \frac{1}{6} \approx 0.267.$$

Akaike, Hirotugu. 1974. “A New Look at the Statistical Model Identification.” *IEEE Transactions on Automatic Control* 19 (6): 716–23. <https://doi.org/10.1109/TAC.1974.1100705>.

Schwarz, Gideon. 1978. “Estimating the Dimension of a Model.” *The Annals of Statistics* 6 (2): 461–64. <https://doi.org/10.1214/aos/1176344136>.